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BUFFER MODULE AND ELECTRONIC DEVICE

Abstract

A buffer module including a main body, a driven component, and a sliding assembly is provided. The driven component is movably disposed on the main body and is used to move relatively to the main body. The sliding assembly is slidably disposed on the main body along a moving direction, where the sliding assembly is used to contact the driven component to suppress the movement of the driven component. In addition, an electronic device including the buffer module is also provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of China application serial no. 202410208709.4, filed on Feb. 23, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates a buffer module and an electronic device; more particularly, the disclosure relates to a buffer module that generates a buffering effect through acceleration and an electronic device including the buffer module.

Description of Related Art

[0003] A server module installed in a chassis of a plug-in server may be connected by plugging into a connector within the chassis. If a user inserts the server module into the chassis at an excessive speed, the connector inside the chassis and the corresponding connector of the server module may collide at high speed, potentially causing damage.

SUMMARY

[0004] The disclosure provides a buffer module and an electronic device that are capable of preventing damage to a connector due to an excessive plugging speed of an electronic assembly.

[0005] According to an embodiment of the disclosure, a buffer module including a main body, a driven component, and a sliding assembly is provided. The driven component is movably disposed on the main body and used to move relatively to the main body. The sliding assembly is slidably disposed on the main body, where the sliding assembly is used to contact the driven component, so as to suppress the movement of the driven component.

[0006] According to an embodiment of the disclosure, an electronic device including a device body, an electronic assembly, and a buffer module is provided. The electronic assembly is detachably installed on the device body. The buffer module is disposed on a surface of the electronic assembly and includes a main body, a driven component, and a sliding assembly. The driven component is movably disposed on the main body. The sliding assembly is slidably disposed on the main body, where the sliding assembly is used to contact the driven component, so as to suppress the movement of the driven component.

[0007] In an embodiment of the disclosure, the sliding assembly includes a sliding component and a counterweight component, the sliding component is slidably disposed on the main body, and the counterweight component is disposed on the sliding component.

[0008] In an embodiment of the disclosure, the sliding assembly further includes an elastic component, and the elastic component is connected between the counterweight component and the main body.

[0009] In an embodiment of the disclosure, the sliding assembly is equipped with a buffer portion, the buffer portion is adjacent to the driven component, and the buffer portion is used to contact the driven component to decelerate the movement of the driven component.

[0010] In an embodiment of the disclosure, the sliding assembly includes a sliding component and a connection rod, the sliding component is slidably disposed on the main body, the connection rod is movably connected to the main body and the sliding component, and the buffer portion is disposed at one end of the connection rod far from the sliding component.

[0011] In an embodiment of the disclosure, the connection rod has a first connection portion and a second connection portion, and the connection rod is connected to the sliding component through the first connection portion and is rotatably connected to the main body through the second connection portion.

[0012] In an embodiment of the disclosure, the sliding assembly is equipped with a stopper, the stopper is adjacent to the driven component, and the stopper is used to contact the driven component to stop the movement of the driven component.

[0013] In an embodiment of the disclosure, the sliding assembly includes a sliding component and a connection rod, the connection rod is movably connected to the main body and the sliding component, and the stopper is disposed at one end of the connection rod far from the sliding component.

[0014] In an embodiment of the disclosure, the driven component is a gear and is rotatably disposed on the main body.

[0015] In an embodiment of the disclosure, the electronic device further includes a rack that is disposed on the device body, the driven component is a gear and is rotatably disposed on the main body, and the gear is engaged with the rack.

[0016] In an embodiment of the disclosure, the electronic device further includes a deceleration module, where the electronic assembly is equipped with a leaning portion, and the electronic assembly is used to contact the deceleration module through the leaning portion to decelerate the electronic assembly.

[0017] In an embodiment of the disclosure, the deceleration module includes a contact component and an elastic component, the contact component is slidably disposed on the device body, the elastic component is connected between the contact component and the device body, and the leaning portion of the electronic assembly is used to push against the contact component.

[0018] According to an embodiment of the disclosure, a buffer module is disposed on an electronic assembly. The electronic assembly is used to move along a moving direction, and the buffer module includes a main body, a driven component, and a sliding assembly. The driven component is movably disposed on the main body and used to move relatively to the main body. The sliding assembly includes a sliding component that is slidably disposed on the main body and used to move relatively to the driven component in a direction opposite to the moving direction and contact the driven component, so as to suppress the movement of the driven component.

[0019] In view of the above, the buffer module, as provided in one or more embodiments of the disclosure, includes the sliding assembly that may generate displacement in response to the acceleration of the electronic assembly. When the sliding assembly generates displacement, the movement of the driven component is suppressed, consequently mitigating the motion of the electronic assembly and achieving a buffering effect. Thus, in light of the above-mentioned operation manner of the buffer module with a straightforward design and a compact structure, when a user forcefully inserts the electronic assembly into the device body by applying an excessive force, leading to excessive acceleration of the electronic assembly, the buffer module may decelerate or halt the movement of the electronic assembly through the buffering effect, thereby preventing damage to the connector due to the excessive plugging speed of the electronic assembly.

[0020] Several exemplary embodiments accompanied with figures are described in detail below to further describe the disclosure in details.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0022] FIG. 1 is a three-dimensional view illustrating some components of an electronic device

according to an embodiment of the disclosure.

[0023] FIG. 2 illustrates the electronic assembly depicted in FIG. 1 is separated from the device body.

[0024] FIG. 3 is a partial enlarge view of the electronic device depicted in FIG. 1.

[0025] FIG. 4 is a partial enlarge view of the electronic assembly depicted in FIG. 2.

[0026] FIG. 5 is an exploded view of the buffer module depicted in FIG. 4.

[0027] FIG. 6 is an exploded view of some components of the electronic device depicted in FIG. 1.

[0028] FIG. 7A to FIG. 7C illustrate a moving process of the buffer module depicted in FIG. 4.

[0029] FIG. 8 is a partial three-dimensional view of the electronic device depicted in FIG. 1.

[0030] FIG. 9A and FIG. 9B illustrate an operation manner of the deceleration module depicted in FIG. 8.

[0031] FIG. 10 is a partial bottom view of an electronic device according to another embodiment of the disclosure.

[0032] FIG. 11A and FIG. 11B illustrate an operation manner of the buffer module depicted in FIG. 10.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

[0033] FIG. 1 is a three-dimensional view illustrating some components of an electronic device according to an embodiment of the disclosure. FIG. 2 illustrates the electronic assembly depicted in FIG. 1 is separated from the device body. FIG. 3 is a partial enlarge view of the electronic device depicted in FIG. 1. With reference to FIG. 1 to FIG. 3, an electronic device **100** provided in this embodiment is, for instance, a server and includes a device body **110** and at least one electronic assembly **120**. The device body **110** is, for instance, a chassis and is equipped with a connector **112**. The electronic assembly **120** is, for instance, a server module and includes a tray **122** and a circuit board module carried on the tray **122**. For the sake of brief and concise illustration, the circuit board module is not shown in the drawings. The electronic assembly **120** is detachably installed on the device body **110**, and the connector **126** of the electronic assembly **120** is used to be inserted into the connector **112** of the device body **110** along a moving direction D.

[0034] FIG. 4 is a partial enlarge view of the electronic assembly depicted in FIG. 2. FIG. 5 is an exploded view of the buffer module depicted in FIG. 4. With reference to FIG. 4 and FIG. 5, the electronic device **100** provided in this embodiment further includes a buffer module **130**. The buffer module **130** is disposed on a surface of the electronic assembly **120** and includes a main body **132**, a driven component **134**, and a sliding assembly **136**. The main body **132** is, for instance, a partial structure of a sidewall **122a** of the tray **122** of the electronic assembly **120**. The driven component **134** is movably disposed on the main body **132** and is used to move relatively to the main body **132** as the main body **132** moves along the moving direction D. The sliding assembly **136** is slidably disposed on the main body **132** along the moving direction D. The sliding assembly **136** is used to move relatively to the main body **132** and the driven component **134** in a direction opposite to the moving direction D as the main body **132** accelerates along the moving direction D, so that the sliding assembly **136** generates displacement towards the driven component **134** and generates a frictional force through contacting the driven component **134** or is engaged with the driven component **134**, thereby suppressing the movement of the driven component **134**.

[0035] As mentioned above, the buffer module **130** provided in this embodiment includes the sliding assembly **136** that may generate displacement as the electronic assembly **120** accelerates. When the sliding assembly **136** generates displacement, the movement of the driven component **134** is suppressed, thereby mitigating the motion of the electronic assembly **120** and achieving a buffering effect. Accordingly, in light of the above-mentioned operation manner of the buffer module **130** with a straightforward design and a compact structure, when a user forcefully inserts the electronic assembly **120** into the device body **110** by applying an excessive force, leading to excessive acceleration of the electronic assembly **120**, the buffer module **130** may decelerate or halt the movement of the electronic assembly **120** through the buffering effect, thereby preventing

damage to the connectors **112** and **126** due to the excessive plugging speed of the electronic assembly **120**.

[0036] FIG. **6** is an exploded view of some components of the electronic device depicted in FIG. **1**. FIG. **7A** to FIG. **7C** illustrate a moving process of the buffer module depicted in FIG. **4**. With reference to FIG. **6** to FIG. **7C**, specifically, the driven component **134** provided in this embodiment is, for instance, a gear and is rotatably disposed on the main body **132**.

Correspondingly, the electronic device **100** further includes a rack **140** fixedly arranged on the device body **110**, and the gear (the driven component **134**) is engaged with the rack **140** and is used to move along the rack **140** as the main body **132** moves along the moving direction D. Thereby, when the sliding assembly **136** suppresses the movement of the driven component **134** (the gear) as described above, the movement of the driven component **134** (the gear) along the rack **140** is decelerated or halted, thus decelerating or stopping the electronic assembly **120** due to the mutual engagement of the driven component **134** (the gear) and the rack **140**. The buffering mechanism of the buffer module **130** provided in this embodiment is further elaborated below.

[0037] With reference to FIG. **4** and FIG. **5**, the sliding assembly **136** provided in this embodiment includes a sliding component **1361**, a counterweight component **1362**, an elastic component **1363**, and a connection rod **1364**. The sliding component **1361** is slidably disposed on the main body **132** along the moving direction D through a cooperation of a sliding groove S of the sliding component **1361** with a column body C (e.g., a bolt) on the main body **132**. The counterweight component **1362** is, for instance, a lead ball and fixedly arranged on the sliding component **1361**. The elastic component **1363** is, for instance, a compression spring and connected between the counterweight component **1362** and the main body **132**. The connection rod **1364** is movably connected to the main body **132** and the sliding component **1361**. Specifically, the connection rod **1364** has a first connection portion **1364a** and a second connection portion **1364b**. The connection rod **1364** is connected to the sliding component **1361** through the first connection portion **1364a**. The first connection portion **1364a** is, for instance, a sliding groove and may cooperate with a protruding column **1361a** of the sliding component **1361**, thus allowing the sliding component **1364** to be slidably disposed between the connection rod **1364** and the main body **132**. The second connection portion **1364b** is, for instance, a pivot hole. The second connection portion **1364b** is located between the first connection portion **1364b** and the driven component **134**. The sliding component **1364** is rotatably connected to the main body **132** through the second connection portion **1364b**. In addition, the sliding assembly **136** has a buffer portion **136a** and a stopper **136b** on the connection rod **1364**. The buffer portion **136a** is, for instance, a rubber or any other elastic buffer material and disposed at one end of the connection rod **1364** far from the sliding component **1364**, and the buffer portion **136a** is adjacent to the driven component **134**. The stopper **136b** is located at the end of the connection rod **1364** far from the sliding component **1364** and adjacent to the driven component **134**. The buffer portion **136a** is disposed adjacent to the stopper **136b**.

[0038] When the electronic assembly **120** has excessive acceleration in the moving direction D, the sliding component **1361** and the counterweight component **1362** resist the elastic force of the elastic component **1363** due to the inertia of the counterweight component **1362** and generate displacement relative to the main body **132** in the direction opposite to the moving direction D as shown in FIG. **7A** to FIG. **7B** or FIG. **7A** to FIG. **7C**. The protruding column **1361a** of the sliding component **136** slides at the first connection portion **1364a** of the connection rod **1364** and drives the second connection portion **1364b** of the connection rod **1364** to pivotally rotate towards the driven component **134**. When the sliding assembly **130** reaches a buffer position shown in FIG. **7B**, the buffer portion **136a** contacts the driven component **134** to decelerate the movement of the driven component **134**. When the sliding assembly **130** reaches a stopper position shown in FIG. **7C**, the stopper **136b** stops the teeth of the driven component **134** (the gear) to halt the movement of the driven component **134**. After the electronic assembly **120** is accordingly stopped, the user may slightly pull the electronic assembly **120** in the direction opposite to the moving direction D to

release the engagement between the stopper **136b** and the driven component **134**, thereby allowing the sliding component **1361** and the counterweight component **1362** to be restored by the elastic force of the elastic component **1363**, and the connection rod **1364** is accordingly restored. At this time, the user may continue to push the electronic assembly **120** along the moving direction D to plug the electronic assembly **120** into the device body **110**.

[0039] Although the elastic force of the elastic component **1363** may cause the stopper **136b** in the state shown in FIG. 7C to be disengaged from the driven component **134** due to the elastic force of the elastic component **1363** as long as the elastic force of the elastic component **1363** is sufficient, it should be mentioned that an excessive elastic force of the elastic component **1363** may lead to an insufficient movement of the counterweight component **1362** relative to the main body **132** by the inertial force. Therefore, in this embodiment, the elastic force of the elastic component **1363** is, for instance, less than an engagement force between the stopper **136b** and the driven component **134**, and it is necessary to disengage the stopper **136b** from the driven component **134** by the force applied by the user as described above. In addition, by selecting the counterweight component **1362** with a specific mass, the counterweight component **1362** may generate a sufficient inertial force when the acceleration of the electronic component **120** exceeds a specific value, so as to resist the buffering effect generated by the elastic force of the elastic component **1363** and the frictional force between the structure. The specific values of the acceleration, the inertial force, the elastic force, the frictional force, and the like are common to people having ordinary skill in the pertinent art and may be calculated and implemented by the people, and thus no further explanation is provided herein, and the specific values should not be construed as limitations in the disclosure.

[0040] In this embodiment, the number of the buffer modules **130** may be two, and the two buffer modules **130** may be located on two opposite sides of the electronic component **120** in a direction perpendicular to the moving direction D. For instance, the two buffer modules **130** may be located on the opposite sidewalls **122a** and **122b** of the tray **122** of the electronic component **120** as shown in FIG. 2. Accordingly, the buffering force provided by the buffer module **130** may be more balanced.

[0041] With reference to FIG. 4 and FIG. 5, the buffer module **130** of this embodiment further includes a cushion pad **131** and a positioning elastic piece **133**. The cushion pad **131** is, for instance, a Mylar sheet and is disposed on the main body **132** and used to contact the gear (the driven component **134**), thus preventing an excessive frictional force between the gear (the driven component **134**) and the main body **132** from hindering the rotation of the gear (the driven component **134**). The positioning elastic piece **133** is disposed on the main body **132** and is used to fix an initial position of the gear (the driven component **134**) through the cooperation of a protruding point **1331** of the positioning elastic piece **133** with a positioning hole H of the gear (the driven component **134**).

[0042] FIG. 8 is a partial three-dimensional view of the electronic device depicted in FIG. 1. With reference to FIG. 6 and FIG. 8, the electronic device **100** provided in this embodiment further includes a deceleration module **150** that is disposed on the device body **110** and located between the rack **140** and the connector **112** (indicated in FIG. 2) in the moving direction D. The electronic assembly **120** is used to contact the deceleration module **150** along the moving direction D and be decelerated by the deceleration module **150**, so as to ensure that the speed at which the electronic assembly **120** is plugged into the device body **110** is not overspeed. Specifically, the deceleration module **150** provided in this embodiment includes a contact component **152** and an elastic component **154**. The contact component **152** is slidably disposed on the device body **110** along the moving direction D, and the elastic component **154** is, for instance, a compression spring connected between the contact component **152** and the device body **110**. As shown in FIG. 4, the tray **122** of the electronic assembly **120** is equipped with a leaning portion **1221**, the leaning portion **1221** is adjacent to the connector **126**, and the leaning portion **1221** is used to contact the contact component **152** of the deceleration module **150** to decelerate the electronic assembly **120**.

[0043] FIG. 9A and FIG. 9B illustrate an operation manner of the deceleration module depicted in FIG. 8. When the user pushes the electronic assembly **120** to a location close to the connector **112**, the electronic assembly **120** pushes against the contact component **152** through the leaning portion **1221** as shown in FIG. 9A to FIG. 9B and resists the elastic force of the elastic component **154**. During this process, the electronic assembly **120** is decelerated by the elastic force of the elastic component **154**. After the electronic assembly **120** stops due to the deceleration effect, the user may push the electronic assembly **120** to the end along the moving direction D to plug the electronic assembly **120** into the device body **110**. When the electronic assembly **120** moves away from the deceleration module **150** in the direction opposite to the moving direction D, the contact component **152** is restored from the location shown in FIG. 9B to the location shown in FIG. 9A due to the elastic force of the elastic component **154**.

[0044] FIG. 10 is a partial bottom view of an electronic device according to another embodiment of the disclosure. FIG. 11A and FIG. 11B illustrate an operation manner of the buffer module depicted in FIG. 10. In an electronic device **200** shown in FIG. 10 to FIG. 11B, the arrangement and the operation manner of a device body **210**, an electronic assembly **220**, a buffer module **230**, a main body **232**, a driven component **234**, a sliding assembly **236**, a buffer portion **236a**, a stopper **236b**, a sliding component **2361**, a counterweight component **2362**, an elastic component **2363**, a rack **240**, and a deceleration module **250** are the same or similar to the arrangement and the operation manner of the device body **110**, the electronic assembly **120**, the buffer module **130**, the main body **132**, the driven component **134**, the sliding assembly **136**, the sliding component **1361**, the counterweight component **1362**, the elastic component **1363**, the rack **140**, and the deceleration module **150** provided in the previous embodiment and thus are not further elaborated hereinafter.

[0045] The main difference between the electronic device **200** and the electronic device **100** lies in that the buffer module **230**, the rack **240**, and the deceleration module **250** in the electronic device **200** are disposed on the underside of the electronic assembly **220** rather than on the left and right sides of the electronic assembly **220**, thus preventing the difficulty in arranging the buffer module **230**, the rack **240**, and the deceleration module **250** due to the insufficient space between the left and right sides of the electronic assembly **220** and the device body **110**. In addition, in the electronic device **200**, the sliding assembly **236** does not include the connection rod **1364** provided in the previous embodiment, and the buffer portion **236a** and the stopper **236b** are directly disposed on one side of the sliding component **2361** and may, as shown in FIG. 11A to FIG. 11B, approach the driven component **234** through the movement of the sliding component **2361**.

[0046] When the electronic assembly **220** has the excessive acceleration in the moving direction D, the sliding component **2361** and the counterweight component **2362** resist the elastic force of the elastic component **2363** through the inertia of the counterweight component **2362** and drive the sliding component **2361** to generate displacement relative to the main body **232** in the direction opposite to the moving direction D, as shown in FIG. 11A to FIG. 11B. The buffer portion **236a** contacts the driven component **234** to decelerate the movement of the driven component **234**. Accordingly, the electronic assembly **220** may be prevented from being plugged into the electronic device **200** at an excessively fast speed.

[0047] To sum up, the buffer module, as provided in one or more embodiments of the disclosure, includes the sliding assembly that may generate displacement in response to the acceleration of the electronic assembly. When the sliding assembly generates displacement, the movement of the driven component is suppressed, consequently mitigating the motion of the electronic assembly and achieving a buffering effect. Thus, in light of the above-mentioned operation manner of the buffer module with a straightforward design and a compact structure, when a user forcefully inserts the electronic assembly into the device body by applying an excessive force, leading to excessive acceleration of the electronic assembly, the buffer module may decelerate or halt the movement of the electronic assembly through the buffering effect, thereby preventing damage to the connector due to the excessive plugging speed of the electronic assembly.

[0048] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. A buffer module, comprising: a main body; a driven component, movably disposed on the main body and moved relatively to the main body; and a sliding assembly, slidably disposed on the main body, wherein the sliding assembly is used to contact the driven component, so as to suppress the movement of the driven component.
2. The buffer module according to claim 1, wherein the sliding assembly comprises a sliding component and a counterweight component, the sliding component is slidably disposed on the main body, and the counterweight component is disposed on the sliding component.
3. The buffer module according to claim 2, wherein the sliding assembly further comprises an elastic component, and the elastic component is connected between the counterweight component and the main body.
4. The buffer module according to claim 1, wherein the sliding assembly is equipped with a buffer portion, the buffer portion is adjacent to the driven component, and the buffer portion is used to contact the driven component to decelerate the movement of the driven component.
5. The buffer module according to claim 4, wherein the sliding assembly comprises a sliding component and a connection rod, the sliding component is slidably disposed on the main body, the connection rod is movably connected to the main body and the sliding component, and the buffer portion is disposed at one end of the connection rod away from the sliding component.
6. The buffer module according to claim 5, wherein the connection rod has a first connection portion and a second connection portion, and the connection rod is connected to the sliding component through the first connection portion and is rotatably connected to the main body through the second connection portion.
7. The buffer module according to claim 1, wherein the sliding assembly is equipped with a stopper, the stopper is adjacent to the driven component, and the stopper is used to contact the driven component to stop the movement of the driven component.
8. The buffer module according to claim 7, wherein the sliding assembly comprises a sliding component and a connection rod, the connection rod is movably connected to the main body and the sliding component, and the stopper is disposed at one end of the connection rod far from the sliding component.
9. The buffer module according to claim 1, wherein the driven component is a gear and is rotatably disposed on the main body.
10. An electronic device, comprising: a device body; an electronic assembly, detachably installed on the device body; and a buffer module, disposed on a surface of the electronic assembly and comprising: a main body; a driven component, movably disposed on the main body; and a sliding assembly, slidably disposed on the main body, wherein the sliding assembly is used to contact the driven component, so as to suppress the movement of the driven component.
11. The electronic device according to claim 10, wherein the sliding assembly comprises a sliding component and a counterweight component, the sliding component is slidably disposed at the main body, and the counterweight component is disposed at the sliding component.
12. The electronic device according to claim 11, wherein the sliding assembly further comprises an elastic component, and the elastic component is connected between the counterweight component and the main body.
13. The electronic device according to claim 10, wherein the sliding assembly is equipped with a buffer portion, the buffer portion is adjacent to the driven component, and the buffer portion is used

to contact the driven component to decelerate the movement of the driven component.

14. The electronic device according to claim 13, wherein the sliding assembly comprises a sliding component and a connection rod, the sliding component is slidably disposed on the main body, the connection rod is movably connected to the main body and the sliding component, and the buffer portion is disposed at one end of the connection rod away from the sliding component.

15. The electronic device according to claim 10, wherein the sliding assembly is equipped with a stopper, the stopper is adjacent to the driven component, and the stopper is used to contact the driven component to stop the movement of the driven component.

16. The electronic device according to claim 15, wherein the sliding assembly comprises a sliding component and a connection rod, the sliding component is slidably disposed on the main body, the connection rod is movably connected to the main body and the sliding component, and the stopper is disposed at one end of the connection rod far from the sliding component.

17. The electronic device according to claim 9, further comprising a rack, wherein the rack is disposed on the device body, the driven component is a gear and is rotatably disposed on the main body, and the gear is engaged with the rack.

18. The electronic device according to claim 9, further comprising a deceleration module, wherein the electronic assembly is equipped with a leaning portion, and the electronic assembly is used to contact the deceleration module through the leaning portion to decelerate the electronic assembly.

19. The electronic device according to claim 18, wherein the deceleration module comprises a contact component and an elastic component, the contact component is slidably disposed on the device body, the elastic component is connected between the contact component and the device body, and the leaning portion of the electronic assembly is used to push against the contact component.

20. A buffer module, disposed on an electronic assembly, the electronic assembly being used to move along a moving direction, the buffer module comprising: a main body; a driven component, movably disposed on the main body and used to move relatively to the main body; and a sliding assembly, comprising: a sliding component, slidably disposed on the main body and used to move relatively to the driven component in a direction opposite to the moving direction and contact the driven component, so as to suppress the movement of the driven component.
